INCH-POUND

MIL-C-24643/10A
14 March 1994
SUPERSEDING
MIL-C-24643/10
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSPBTMU (INCLUDING VARIATION LSPBTM)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - One copper and one constant conductor for each pair, uncoated. AWG size 22, class K stranding. The conductor shall have a cross-sectional area within plus or minus 4 percent of the nominal circular mils as specified in MIL-C-24643. Temperature/electromotive force relationship shall follow ANSI MC96.1. Any necessary joints in the individual strands shall be silver-soldered for copper, and welded for constantan.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.015 inch. Telephone identification code applied by method 3.

Fourth - Each pair shall be one copper and one constantan conductor cabled with a lay not greater than 3 inches.

Fifth - The specified number of pairs (see table I) shall be cabled with a lay not greater than 10 inches in the outer layer. Fillers shall be used in all valleys as necessary to form a round cable. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward.

Sixth - Binder tape.

AMSC N/A FSC 6145

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Construction (watertight) (continued)

Construction variant for type LSPBTMU

* Seventh- Cross-linked polyolefin jacket.

Construction variant for type LSPBTMU

Seventh- Cross-linked polyolefin jacket.
* Eighth - Braided metal armor.

TABLE I. Details

Military 1/ part no. M24643/10	Size2/	Number of pairs	Overall Diameter (inches)				Cold bending diameter maximum (inches)
			Type LSPBTMU		Type LSPBTM		
			min	max	min	max	
-01UN	5	5	0.499	0.540	0.549	0.590	8
-02UN	15	15	.694	.750	.744	.800	10
-03UN	30	30	.907	.980	.957	1.030	13

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSPBTMU and LSPBTM.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
ANSI MC96.1 calibration limits of error	±1°C or ±0.75 percent
Voltage withstand - volts, root mean square, minimum	1000
* Insulation resistance - conductor to conductor, megohms/1000 feet, minimum	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
* **Group B:**	
Cold bending, cable - (see table I for mandrel diameter)	No damage
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35

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EXAMINATION AND TESTS (continued):

	Requirements
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
* Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged minimum	50
Shrinkage	No failure
Permanence of printing for type LSPBTMU only (jacket) cycles, minimum	125
Heat distortion - percent of unaged, maximum	30
* Cable sealant removability	No failure
Armor - conformance to material construction and coverage (type LSPBTM only)	No failure
Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION: Requirements

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket/insulation)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH:

Size	Feet (nominal)
5	2500
15	1500
30	1000

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project No. 6145-2057-10)

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